CLASS: Deep Partial Label Feature Selection with Cluster-Guided Disambiguation and Structured Sparsity

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Abstract

Partial label learning efficiently extracts accurate labels from weak supervision, where each instance has multiple candidate labels but only one is correct. Existing partial label learning methods typically employ a two-stage process and fall into a suboptimal solution in high-dimensional settings, lacking an effective feedback mechanism between feature selection and label disambiguation. Nevertheless, existing methods suffer from at least one of two issues, i.e., redundant features and the uncertainty of label disambiguation. To address these problems, we propose a unified and deep partial label feature selection method with *CLuster-guided disAmbiguation and Structured Sparsity* (CLASS), which simultaneously preserve discriminative features and promotes candidate label disambiguation by bidirectional optimization. Specifically, we integrate nonlinear deep networks to capture high-order semantic relations and feature interactions, while employing a linear sparse gating mechanism to preserve the label-specific, discriminative and interpretable features. Moreover, a cluster-guided disambiguation module is designed to enforce inter-class global separation and intra-class local cohesion via distance constraint and confidence penalty and dynamically align predicted labels with the semantics of class prototypes. Extensive experimental results validate the superiority and effectiveness of our proposed CLASS over state-of-the-art methods.

CCS Concepts

• **Computing methodologies** → **Supervised learning by classification**; **Feature selection**; **Neural networks**.

Keywords

Partial Label Learning, Deep Feature Selection, Cluster-guided Disambiguation, Weakly Supervised Learning

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1 Introduction

Partial label learning (PLL) [34] refers to a weakly supervised learning paradigm where each training instance is associated with a set of candidate labels, among which only one is correct, but the ground-truth label is not explicitly known. Disambiguation is one of the pivotal challenges in PLL. Figure 1(a) shows a classical PLL scenario in automatic face naming, in which given the news "James talks about Doncic", James, Doncic, and Davis are candidate labels, and the task is to identify the ground-truth one. As an important emerging research field in weakly supervised problems [18, 51], PLL methods [5, 6, 37–39, 43] has been widely applied in multiple practical fields, such as computer vision [45, 48], natural language processing [31, 32] etc. Therefore, numerous PLL methods with promising performance have been proposed [10, 12, 41, 44].

In literature, as stated in [34], the main strategies of existing PLL methods for reducing label disambiguation can be classified into two main categories, including the average-based disambiguation strategy [7, 9, 20, 47] and the identification-based disambiguation strategy [4, 15, 23]. The average-based disambiguation strategy treats all the candidate labels of an example in an equal manner, i.e., convex learning from partial labels (CLPL) [7], an instance-based approach (IPAL) [47] and generalized large margin kNN for partial label learning (GLMNN-PLL) [9]. The identification-based disambiguation strategy identifies the ground-truth label along with model training, i.e., complementary classifier induced partial label learning (PL-CL) [15], an LM partial label machine (LM-PLANE) [4] and graph matching based partial label learning (GM-PLL) [23]. In addition, real-world data increasingly exhibits high dimensionality, particularly in video semantic recognition and bioinformatics, where direct processing of high-dimensional data is not only computationally inefficient, but the noise and redundant

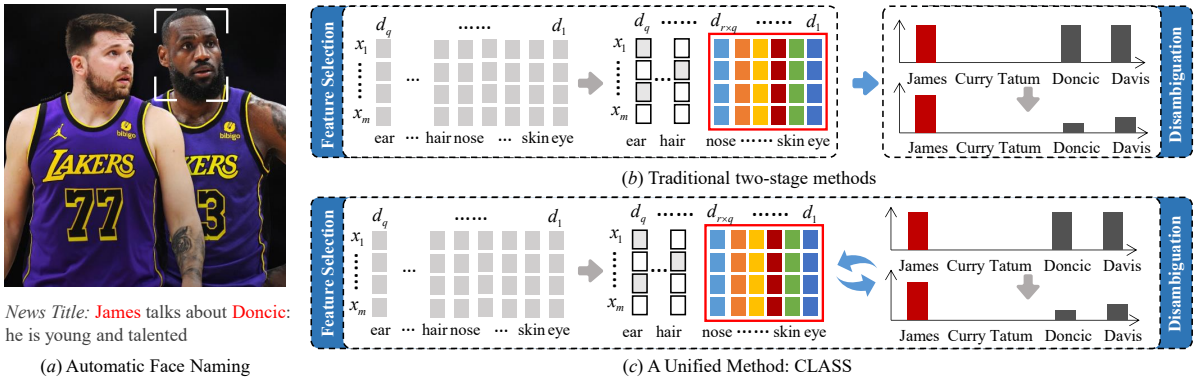

Figure 1: From Two-Stage to Deep Unified Methods: Partial Label Feature Learning for High-Dimensional Scenarios.

features also degrade label disambiguation performance. In this context, a variety of partial label feature learning methods have been widely proposed, falling primarily into two categories: partial label feature dimensionality reduction [1, 46] and partial label feature selection [2, 30, 33]. Compared to dimensionality reduction, feature selection methods [17–19] aim to select discriminative feature subsets directly from the original features and preserve the superior interpretability of features.

However, existing partial label disambiguation strategies and partial label feature selection methods each have their limitations: the former either interfere with the ground-truth label by treating all candidate labels equally, or struggle to stably identify the ground-truth label when candidates are similar, failing to fully resolve label uncertainty; the latter, adopting a two-stage pipeline, as shown in Figure 1(b), fall into a suboptimal solution in high-dimensional settings, lacking an effective feedback mechanism between feature selection and label disambiguation. As a result, they face two critical challenges: (1) redundant features may mislead the disambiguation process; (2) conversely, biases in pseudo-labels can undermine the stability of feature selection. When these two components are optimized independently, the learning model fails to accurately capture the true relationship between key features and labels, ultimately converging to a suboptimal state due to their mutual interference and severely limiting its applications.

To solve these problems, we propose a unified deep learning model called CLASS by integrating feature selection and label disambiguation. Specifically, to capture high-order semantic relations and complex feature interactions under partial supervision, we design a nonlinear deep network via a linear structured sparse gating unit to extract more discriminative and interpretable representation. To effectively model the latent semantic structure of candidate labels, we introduce a disambiguation strategy by integrating a distance constraint with a confidence penalty. Besides, its cluster-guided disambiguation module enforces inter-class global separation and intra-class local cohesion to align predicted labels with the semantics of class prototypes dynamically. In this way, CLASS jointly embeds feature selection and label disambiguation in a single deep objective, enabling bidirectional refinement of features and label confidence. Efficient iterative optimization algorithms are designed to solve the proposed CLASS. Extensive experimental results on various benchmarks demonstrate the superiority and effectiveness of the proposed CLASS.

Our main contributions are summarized as follows:

- We propose the deep method CLASS to solve this crucial, but rarely studied problem from partial label data. To our knowledge, this may be the first attempt to simultaneously tackle feature selection and label disambiguation.
- Deep feature selection network with structured sparsity is designed to explore the high-order interactions between features and preserve their more discriminative information.
- We design cluster-guided disambiguation mechanism to implement global separation and local cohesion via distance constraint and confidence penalty to dynamically align predicted true labels with semantic of class prototypes.
- Extensive experimental results on diverse datasets demonstrate that our CLASS consistently outperforms other approaches and confirm its effectiveness and robustness.

2 Related Work

2.1 Partial Label Learning

Disambiguation is important in partial label learning research [24, 25, 40]. Early representative methods, such as PL-KNN [14] and PL-SVM [26], primarily rely on simple averaging strategies or confidence-based iterative optimization for label disambiguation. Subsequently, some mid-stage approaches moved beyond naive averaging and local propagation. For instance, Zhang et al. [8, 36, 49] proposed PL-ECOC, PL-AGGD and SURE based on structured output codes, graph-based constraints, and ℓ_∞-norm regularization, respectively. These approaches focused instead on structural modeling among candidate labels and objective design. As data scale and task complexity continue to grow, traditional shallow PLL methods have shown limitations in representation capacity and feature modeling, making them less suitable for high-dimensional, heterogeneous, or unstructured data. In recent years, an increasing number of studies have integrated PLL with deep neural networks, advancing the field toward greater generalization. Lv et al. [21, 22, 42] focus on aspects like model-agnostic risk estimation, robustness analysis of average-based strategies, and leveraging weighted loss, bringing the field closer to deep learning paradigms. However, most of these

methods focus on the design of disambiguation mechanisms while overlooking the impact of feature representation quality on disambiguation, thereby limiting the overall classification performance.

2.2 Partial Label Feature Learning

PLL often involves high-dimensional data such as image features or text embeddings, where irrelevant or noisy features may obscure the true label. Although classical feature selection methods such as Fisher Score [11], Mutual Information [29], and Chi-Square Test [16] are widely used, their performance degrades in PLL settings due to the lack of supervision signal. Recent efforts have extended feature learning to partial label scenarios, which can be broadly divided into feature dimensionality reduction and feature selection. Feature dimensionality reduction methods aim to learn low-dimensional representations that facilitate disambiguation. For instance, CENDA [1] realizes confidence-aware dimensionality reduction by maximizing projected features-label confidence dependency via the Hilbert-Schmidt Independence Criterion. DRAW [46] introduces an adaptive mechanism, shifting from PCA to LDA with improved label confidence. These effective methods lack sparsity control and interpretability compared to feature selection approaches. In contrast, feature selection methods identify task-relevant features to enhance generalization and interpretability. SAUTE [2] addresses partial label selection via mutual-information-based dependency maximization and submodular optimization refinement. RFUTE [33] adopts random forest disambiguation followed by entropy-based ranking to mitigate class imbalance. PFNL [30] uses embedding with manifold regularization in feature and label spaces to suppress pseudo-label noise. Although these methods have been applied to PLL scenarios, most use a two-stage pipeline that separates disambiguation and feature selection, potentially overlooking their mutual dependencies and limiting performance gains.

3 Methodology

3.1 The proposed CLASS method

Existing partial label feature selection methods often sequentially decouple feature selection and label disambiguation, creating a vicious circle: ambiguous labels admit noisy features, while poor features degrade disambiguation, trapping performance in a local minimum. To address these issues, we propose CLASS, which unifies deep feature selection and label disambiguation into a framework, whose main structure is shown in Figure 2. CLASS mainly includes two modules: (1) deep feature selection network that combines a nonlinear deep structure with a linear sparse gating mechanism, capable of effectively extracting discriminative features while automatically filtering out redundant or irrelevant information; (2) cluster-guided disambiguation module is designed to enforce inter-class global separation and intra-class local cohesion via distance constraints and confidence penalties and dynamically align predicted labels with the semantics of class prototypes.

Deep feature selection network with linear sparse gate. To address the limitation that existing feature selection methods relying mostly on a single mechanism, fail to balance sparsity, discriminability, and dynamic adaptability, we design deep feature selection module. Denote the ground-truth label $\mathbf{y}_i^* \in \mathbf{S}_i$ for each instance $\mathbf{x}_i \in \mathcal{X}$. The prediction model $g_{\mathbf{\Theta},\mathbf{W}}(\mathbf{x}_i)$, which consists of a linear sparse selector and a nonlinear deep network component, and is expressed as residual feed-forward neural networks [13]:

$$g_{\mathbf{\Theta},\mathbf{W}}(\mathbf{x}_i) = \mathbf{\Theta}^T \mathbf{x}_i + h_{\mathbf{W}}^{\mathrm{NN}}(\mathbf{x}_i),$$

where the linear term $\mathbf{\Theta}^T \mathbf{x}_i$ enables feature selection through sparsity control, promoting interpretability. The nonlinear term $h_{\mathbf{W}}^{\mathrm{NN}}(\mathbf{x}_i)$, modelled by a feed-forward neural network with parameters $\mathbf{W}$, enhances the model's capacity to capture complex feature interactions. Given $\mathbf{\Theta}$, motivated by [27], the contribution of j-th feature for classification can be measured by $\|\mathbf{\Theta}_j\|_2 = \sqrt{\sum_k \mathbf{\Theta}_{j,k}^2}$, where larger values signify higher importance. Besides, the number of selected features is inversely controlled by λ such that increasing λ reduces the number of selected features. In the presence of noisy or ambiguous candidate labels, the strictly convex squared error term guides gradient descent to converge to the true data manifold, mitigating label ambiguity effects during optimization. Therefore, the objective of deep feature selection network with structured sparsity regularization can be formulated as:

$$\begin{cases} \min\limits_{\mathbf{\Theta},\mathbf{W}} \sum\limits_{i=1}^{m} \left\| g_{\mathbf{\Theta},\mathbf{W}}(\mathbf{x}_i) - \mathbf{f}_i \right\|^2 + \lambda \|\mathbf{\Theta}\|_{2,1}, \\ \text{s.t.} \quad \|\mathbf{W}_j^{(1)}\|_\infty \le M \|\mathbf{\Theta}_j\|, \ j \in [d], \end{cases} \tag{1}$$

where M and λ are two non-negative parameters to regulate the consistency between the linear selector $\mathbf{\Theta}$ and the nonlinear network weights $W^{(1)}$, and balance the sparsity of features.

Noting that the constraint in Eq. (1) establishes a rigorous linear sparse gating mechanism where the sparsity of linear parameter $\mathbf{\Theta}_j$ directly dictates the activation of nonlinear weight $\mathbf{W}_j^{(1)}$. The corresponding nonlinear transformation pathway is enabled only when $\mathbf{\Theta}_j \neq 0$, yielding a directed sparse response to critical task dimensions. The scaling constant M constrains the ratio between $\mathbf{W}_j^{(1)}$ and $\mathbf{\Theta}_j$, ensuring that the parameters of the deep nonlinear part conform to the feature selection results of the linear part.

Cluster-guided disambiguation. To overcome the prevailing limitation of existing PLL methods that rely solely on the internal associations of candidate label sets, we introduce a cluster-guided disambiguation module via distance constraints and confidence penalties. Concretely, to explicitly leverage global class structures, we introduce cluster-guided prototypes $\{\mathbf{v}_c\}$ as the representative of each class. By analysis, the smaller distance between the sample $\mathbf{x}_i$ and the prototypes $\mathbf{v}_c$ of class c, the confidence f_{ic} is higher than other classes. To align predicted labels with the semantics of class prototypes of partial data, we use f_{ic} as the weight to balance its contribution and enforce inter-class global separation and intra-class local cohesion, whose objective can be formulated as

$$\min_{\mathbf{F}} \sum_{i,c} f_{ic} \|\mathbf{x}_i - \mathbf{v}_c\|^2, \text{s.t. } f_{ic} \ge 0, i \in [m], c \in [q], \tag{2}$$

where $f_{ic} \in [0, 1]$ denotes the confidence of sample i to class c.

The distance constraints have trivial solutions. To address this issue, we introduce confidence penalty that suppresses such degeneration by encouraging higher confidence for instances located

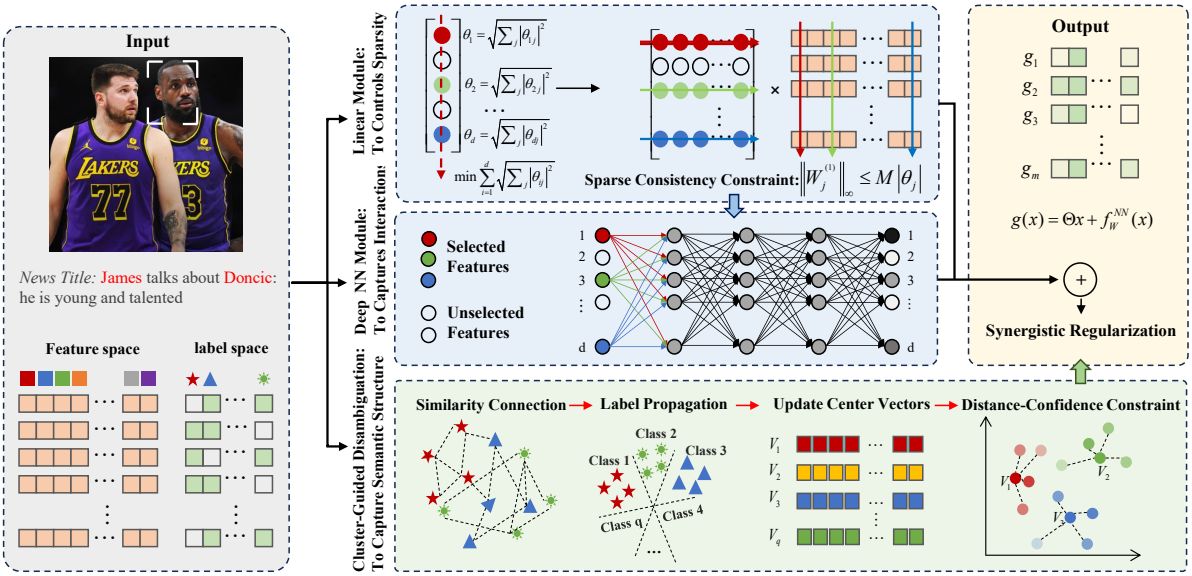

Figure 2: Illustration of the main framework of our proposed CLASS.

near class centers. Therefore, Eq. (2) can be rewritten as:

$$\begin{cases} \min\limits_{\mathbf{F}} \sum\limits_{i,c} f_{ic} \|\mathbf{x}_i - \mathbf{v}_c\|^2 + \mu \cdot \mathbb{I}\left(|\mathbf{x}_i - \mathbf{v}c|^2 \leq \delta\right) \cdot (1 - f_{ic}) \\ \text{s.t. } \mathbf{F}\mathbf{1}_c = \mathbf{1}, f_{ic} \geq 0, i \in [m], c \in [q], \end{cases} \tag{3}$$

where μ is a trade-off parameter, δ is the distance threshold, and $\mathbb{I}(\cdot)$ denotes the indicator function. However, $\mathbb{I}(\cdot)$ is discrete and hard to optimize, therefore we use a smooth approximation $\exp(\cdot)$ instead. The problem (3) can be rewritten as:

$$\begin{cases} \min\limits_{\mathbf{F}} \sum\limits_{i,c} f_{ic} ||\mathbf{x}_i - \mathbf{v}_c||^2 + \mu \sum\limits_{i,c} \exp\left(-\frac{||\mathbf{x}_i - \mathbf{v}_c||_2^2}{\delta}\right)(1 - f_{ic}) \\ \text{s.t. } \mathbf{F}\mathbf{1}_c = \mathbf{1}, f_{ic} \geq 0, i \in [m], c \in [q]. \end{cases} \tag{4}$$

Finally, by incorporating Eq. (1) with Eq. (4), the final objective function of our proposed CLASS is formulated as:

$$\begin{cases} \min\limits_{\mathbf{\Theta},\mathbf{W},\mathbf{F}} \mathcal{L} = \sum\limits_{i} \left|\left|g_{\mathbf{\Theta},\mathbf{W}}(\mathbf{x}_i) - \mathbf{f}_i\right|\right|^2 + \lambda ||\mathbf{\Theta}||_{2,1} \\ \quad + \sum\limits_{i,c} \left[f_{ic} \left|\left|\mathbf{x}_i - \mathbf{v}_c\right|\right|^2 + \mu \exp\left(-\frac{\left|\left|\mathbf{x}_i - \mathbf{v}_c\right|\right|_2^2}{\delta}\right)(1 - f_{ic}) \right] \\ \text{s.t. } ||\mathbf{W}_j^{(1)}||_\infty \leq M||\mathbf{\Theta}_j||, \ \mathbf{F}\mathbf{1}_c = \mathbf{1}, f_{ic} \geq 0, i \in [m], c \in [q]. \end{cases} \tag{5}$$

3.2 Alternative Optimization

As described previously, the optimization problem (5) involves three variables with different regularization and constraints, making it difficult to solve directly. In this section, this problem will be solved using an alternating optimization algorithm. Specifically, the variables $\mathbf{\Theta}$, $\mathbf{W}$, and $\mathbf{F}$ are updated in an iterative manner, where each variable is optimized while keeping the others fixed.

(1) Update F: When $\mathbf{\Theta}$ and $\mathbf{W}$ are fixed, the optimization problem (5) decomposes into:

$$\begin{cases} \min\limits_{\mathbf{F}} \sum\limits_{i,c} (g_{ic} - f_{ic})^2 + f_{ic} d_{ic} \\ \text{s.t. } \mathbf{F}\mathbf{1}_c = \mathbf{1}, f_{ic} \geq 0, i \in [m], c \in [q], \end{cases} \tag{6}$$

where $d_{ic} = \|\mathbf{x}_i - \mathbf{v}_c\|_2^2 - \mu \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{v}_c\|^2}{\delta}\right)$ and g_{ic} denotes the c-th output of $g_{\mathbf{\Theta},\mathbf{W}}(\mathbf{x}_i)$. Define $\psi_{ic} = d_{ic} - 2g_{ic}$ and omit the constant term, Eq. (6) is simplified into a quadratic form:

$$\begin{cases} \min\limits_{\mathbf{F}} \sum\limits_{i=1}^{m} \left\|f_{ic} + \frac{1}{2}\psi_{ic}\right\|^2 \\ \text{s.t. } \mathbf{F}\mathbf{1}_c = \mathbf{1}, f_{ic} \geq 0, i \in [m], c \in [q]. \end{cases} \tag{7}$$

Note that the problem (7) is independent between different i, so we can solve the following problem individually for each i:

$$\begin{cases} \min\limits_{\mathbf{f}_i} \|\mathbf{f}_i + \frac{1}{2}\psi_i\|_2^2 \\ \text{s.t. } \mathbf{f}_i \geq 0, \mathbf{f}_i\mathbf{1} = 1. \end{cases} \tag{8}$$

For each i, the Lagrangian function of problem (8) is:

$$\mathcal{L}(\mathbf{f}_i, \alpha, \boldsymbol{\beta}_i) = \|\mathbf{f}_i + \frac{1}{2}\boldsymbol{\psi}_i\|^2 - \alpha(\mathbf{f}_i^\top \mathbf{1} - 1) - \boldsymbol{\beta}_i^\top \mathbf{f}_i, \tag{9}$$

where $\boldsymbol{\psi}_i = [\psi_{i1}, \ldots, \psi_{iq}]^\top$, $\alpha \in \mathbb{R}$ and $\boldsymbol{\beta}_i \in \mathbb{R}^q$, $\boldsymbol{\beta}_i \geq \mathbf{0}$ are Lagrange multipliers. To solve the problem (9), according to the KKT conditions [28], the optimal solution to f_{ic} is:

$$f_{ic} = \left(-\frac{1}{2}\psi_{ic} + \frac{\alpha}{2}\right)_+, \tag{10}$$

substituting Eq.(10) into the constraint $\sum_{c=1}^{q} f_{ic} = 1$, we can obtain:

$$\sum_{c=1}^{q} \left(-\frac{1}{2}\psi_{ic} + \frac{\alpha}{2}\right) = 1 \implies \alpha = \frac{2}{q} + \frac{1}{q}\sum_{c=1}^{q} \psi_{ic}.$$

Note that the detailed optimization process of problem (8) is listed in the Appendix B.1. Once $\mathbf{F}$ is updated, the class centers $\mathbf{v}_c$ are subsequently updated by the procedure in Appendix B.2.2.

(2) Update Θ: With $\mathbf{F}$ and $\mathbf{W}$ fixed, we can rewrite the optimization problem (5) as follows:

$$\min_{\Theta} \sum_i \left\|\Theta^\top \mathbf{x}_i - \mathbf{f}_i\right\|^2 + \lambda \|\Theta\|_{2,1}, \text{ s.t. } \|\mathbf{w}_j^{(1)}\|_\infty \le M\|\Theta_j\| \quad (11)$$

where $\|\Theta\|_{2,1} = \sum_j \|\Theta_{:,j}\|_2$. The subgradient $\mathbf{G} \in \mathbb{R}^{q\times d}$ of $\|\Theta\|_{2,1}$ is computed as:

$$\mathbf{G}_j = \frac{\partial \|\Theta\|_{2,1}}{\partial \Theta_{:,j}} = \begin{cases} \frac{\Theta_{:,j}}{\|\Theta_{:,j}\|_2}, & \Theta_{:,j} \neq 0 \\ 0, & \Theta_{:,j} = 0. \end{cases}$$

Let $\Theta^{(t)}$ denote the value of Θ at the t-th iteration, and $\mathbf{G}^{(t)}$ represent the subgradient of $\|\Theta^{(t)}\|_{2,1}$. For the $(t+1)$-th iteration, the gradient update rule of Eq.(11) is given by:

$$\tilde{\Theta}^{(t+1)} = \Theta^{(t)} - \eta \cdot \left(2\sum_i (\Theta^{(t)\top}\mathbf{x}_i - \mathbf{f}_i)\mathbf{x}_i^\top + \lambda \mathbf{G}^{(t)}\right), \quad (12)$$

where η is the learning rate. Motivated by [3], based on the gradient update rule, we adopt a soft-thresholding operator to project $\tilde{\Theta}^{(t+1)}$ on the constraints in Eq. (11) and obtain its optimal solution:

$$\Theta_j^{(t+1)} = \text{sign}\left(\tilde{\Theta}_j^{(t+1)}\right) \cdot \max\left(\|\tilde{\Theta}_j^{(t+1)}\| - \lambda, 0\right) \quad (13)$$

(3) Update W: With $\mathbf{F}$ and Θ fixed, the problem (5) can be reformulated as the following problem:

$$\min_{W} \sum_i \|g_{\Theta,W}(\mathbf{x}_i) - \mathbf{f}_i\|^2, \text{ s.t. } \|\mathbf{w}_j^{(1)}\|_\infty \le M\|\Theta_j\|. \quad (14)$$

Let $\mathbf{W}^{(t)}$ denote the value of $\mathbf{W}$ at the t-th iteration. Based on the objective function defined in Eq. (14), the gradient update rule for $\mathbf{W}$ in the $(t+1)$-th iteration is given by:

$$\tilde{W}^{(t+1)} = W^{(t)} - \eta \cdot \frac{\partial}{\partial W^{(t)}} \sum_i \|g_{\Theta^{(t+1)},W^{(t)}}(\mathbf{x}_i) - \mathbf{f}_i\|^2, \quad (15)$$

where η is the learning rate. To enforce the structural constraint $\|\mathbf{w}_j^{(1)}\|_\infty \le M\|\Theta_j\|$ specified in Eq. (14), we apply a projection operation on the intermediate update $\tilde{W}^{(t+1)}$ in Eq. (15):

$$\mathbf{w}_j^{(t+1)} = \min\left(1, \frac{M\|\Theta_j\|}{\|\tilde{\mathbf{w}}_j^{(t+1)}\|_\infty}\right) \cdot \tilde{\mathbf{w}}_j^{(t+1)}. \quad (16)$$

Noting that through experimental validation, we found that initialization plays a crucial role in ensuring optimization stability. We first construct a similarity graph $\mathbf{S}$ using a kNN Gaussian kernel, and derive an initial label confidence matrix $\mathbf{F}^{(0)}$ by averaging over candidate labels. Following label propagation [35, 50], $\mathbf{F}^{(t)} = \gamma\mathbf{H}^{(t-1)} + (1-\gamma)\mathbf{F}^{(0)}$, where $\mathbf{H}$ is normalized S, we refine $\mathbf{F}$ iteratively and normalize it to obtain $\hat{\mathbf{F}}$ for confident pseudo-labels. These are then used to initialize class center $\mathbf{v_c}$ as the mean of assigned samples. The linear weight Θ is initialize as all ones to support sparse optimization, while the nonlinear weight $\mathbf{W}^{(1)}$ is initialized via backpropagation under the constraint of Θ. More implementation details are present in Appendix B.2.

Algorithm 1 Optimization Procedure of CLASS

Input: Feature vectors $\mathcal{X} = \mathbb{R}^d$, candidate label sets $\{S_i \subseteq \mathcal{Y}\}_{i=1}^m$
Output: Predicted labels $\hat{\mathbf{y}}_i \in \mathcal{Y}$

```
Initialize confidence matrix F based on label propagation by Appendix B.2.1
Initialize class centers v_c by Appendix B.2.2; feature selector Θ and network weights W by Appendix B.2.3
while not converged and t < T_max do
    Update Θ by Eq. (13)
    Update W by Eq. (16)
    Update confidence matrix F by Eq. (10)
    Update class centers v_c ← Σ_i f_ic x_i / Σ_i f_ic for each c
end while
Predict labels: ŷ_i ← arg max_c g_{Θ,W}(x_i)
return {ŷ_i}, F, Θ, {v_c}
```

The optimization of CLASS is outlined in Algorithm 1. It starts by initializing the confidence matrix $\mathbf{F}$ via label propagation, along with class centers $\mathbf{v}_c$, selector Θ, and network weights $\mathbf{W}$. Then, Θ and $\mathbf{W}$ are alternately updated to select informative features and refine $\mathbf{F}$ and $\mathbf{v}_c$ iteratively until convergence. Final predictions $\hat{\mathbf{y}}_i$ are obtained from $g_{\Theta,W}(\mathbf{x}_i)$.

4 Experiment

4.1 Experimental Settings

Datasets: We conduct the experiments on six partial label datasets, including Lost, MSRCv2, BirdSong, SoccerPlayer, Mirflickr and Yahoo!News, and four synthetic datasets,including FMEAT, HAR, Isolet and Dermatology. More details are listed in Appendix C.1.

Baselines: To validate the effectiveness of **CLASS**, we compare it with four partial label classification methods, including **PL-ECOC** [49], **PL-AGGD** [36], **PRODEN** [21]; and four feature selection methods, including **PFNL** [30], **RFUTE** [33], **SAUTE** [2], and **CENDA** [1]. Detailed descriptions of these methods are provided in Appendix C.2. All experiments were conducted on a laptop equipped with an NVIDIA GeForce RTX 3050 GPU and an Intel 7-core processor. And all the parameter settings in the comparative experiments were set as in the original text.

Parameter Setting: In CLASS, threee regularization parameters M, λ and μ are introduced in the model optimization stage, which are tuned in $\{0.01, 0.1, 1, 10, 100\}$. Datasets are split 1:1 for training and testing. Each experiment is independently conducted ten times, and the final average results with standard deviation of all compared methods are reported.

4.2 Model Classification Performance

To comprehensively evaluate performance of our proposed method, we conduct two groups of experiments. The first group aims to evaluate the model's practical applicability on several real-world and public datasets. The second group focuses on assessing the model's robustness under different numbers of false positive labels.

Practical applicability on real-world datasets: To evaluate the model's practical applicability of our CLASS, we compare it with all of chosen baselines on real-world partial label datasets, i.e.

Table 1: Classification Accuracy and AUC with standard deviation on real-world partial label datasets

Method	Lost		MSRCv2		BirdSong		SoccerPlayer		Mirflickr		Yahoo!News	
	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
PL-ECOC	70.17 ± 1.53	79.10 ± 1.92	50.47 ± 3.21	70.23 ± 2.24	74.04 ± 1.65	81.67 ± 1.74	52.92 ± 2.34	73.11 ± 2.38	60.25 ± 2.07	77.50 ± 2.15	59.94 ± 1.58	75.28 ± 1.96
PL-ECOC-PFNL	18.90 ± 3.54	60.84 ± 2.80	29.53 ± 2.63	63.41 ± 2.16	43.47 ± 2.28	69.18 ± 1.88	22.88 ± 1.67	65.35 ± 1.94	57.12 ± 2.84	74.32 ± 2.43	30.74 ± 1.24	68.03 ± 1.71
PL-ECOC-RFUTE	17.64 ± 2.95	58.21 ± 2.45	36.35 ± 3.63	66.38 ± 2.65	43.89 ± 2.24	70.10 ± 2.05	23.25 ± 4.68	66.29 ± 3.01	37.10 ± 2.36	70.11 ± 2.31	31.89 ± 1.29	69.90 ± 1.64
PL-ECOC-SAUTE	17.62 ± 5.52	59.00 ± 3.73	12.11 ± 3.34	58.90 ± 2.80	33.88 ± 2.14	66.45 ± 2.23	20.21 ± 2.25	63.12 ± 2.54	54.45 ± 3.85	72.77 ± 3.01	27.75 ± 1.61	65.05 ± 1.89
PL-ECOC-CENDA	18.87 ± 3.27	60.91 ± 2.34	48.41 ± 3.78	69.04 ± 3.02	43.35 ± 1.82	70.03 ± 2.17	30.97 ± 2.18	67.86 ± 2.27	19.46 ± 0.34	61.88 ± 1.56	33.33 ± 5.02	70.88 ± 2.84
PRODEN	68.42 ± 3.14	84.31 ± 2.24	49.83 ± 0.74	73.28 ± 1.73	69.88 ± 0.81	83.15 ± 1.60	53.09 ± 0.73	77.89 ± 1.88	66.21 ± 0.13	84.20 ± 1.24	65.34 ± 2.15	84.11 ± 1.73
PRODEN-PFNL	34.72 ± 3.58	65.92 ± 3.01	33.29 ± 1.27	67.70 ± 1.96	44.92 ± 3.51	71.80 ± 2.22	47.49 ± 1.34	73.10 ± 1.69	33.28 ± 3.16	70.64 ± 2.60	42.70 ± 1.64	74.52 ± 2.10
PRODEN-RFUTE	53.58 ± 4.41	76.45 ± 2.80	45.55 ± 1.47	71.21 ± 1.84	40.28 ± 1.18	74.04 ± 2.30	31.95 ± 2.28	68.66 ± 2.10	52.11 ± 2.85	77.09 ± 2.35	48.32 ± 4.34	77.01 ± 2.41
PRODEN-SAUTE	66.25 ± 1.29	82.01 ± 1.95	51.90 ± 2.24	74.48 ± 2.14	58.77 ± 1.83	78.59 ± 2.11	48.75 ± 2.29	75.13 ± 2.22	63.22 ± 2.36	82.20 ± 1.97	58.45 ± 3.34	81.00 ± 2.32
PRODEN-CENDA	63.61 ± 2.83	80.44 ± 2.10	49.36 ± 2.91	73.55 ± 2.40	56.72 ± 3.21	77.40 ± 2.52	54.21 ± 2.42	78.01 ± 2.41	62.45 ± 1.92	81.33 ± 1.82	66.28 ± 2.71	83.35 ± 2.05
PL-AGGD	75.99 ± 2.13	87.20 ± 1.68	46.49 ± 1.42	71.32 ± 1.51	71.78 ± 0.89	84.50 ± 1.55	52.41 ± 0.52	76.60 ± 1.41	65.26 ± 1.83	83.89 ± 1.80	61.34 ± 0.92	81.92 ± 1.64
PL-AGGD-PFNL	42.84 ± 3.57	68.92 ± 2.84	18.49 ± 7.34	60.03 ± 4.60	27.43 ± 1.42	63.44 ± 2.14	24.23 ± 4.21	66.27 ± 3.55	28.42 ± 2.64	66.80 ± 2.41	13.72 ± 3.32	61.13 ± 3.18
PL-AGGD-RFUTE	33.79 ± 2.65	66.13 ± 2.95	27.45 ± 3.83	65.12 ± 3.10	58.95 ± 3.42	78.20 ± 3.02	35.72 ± 4.53	69.22 ± 2.96	28.05 ± 1.54	65.10 ± 1.85	37.22 ± 12.51	73.50 ± 4.01
PL-AGGD-SAUTE	69.40 ± 3.52	85.03 ± 2.21	45.26 ± 2.87	70.87 ± 2.91	67.77 ± 1.48	81.33 ± 1.96	44.32 ± 2.46	73.15 ± 2.45	58.22 ± 0.82	79.10 ± 1.63	49.27 ± 2.95	77.90 ± 2.42
PL-AGGD-CENDA	63.55 ± 2.24	80.52 ± 2.01	40.88 ± 2.97	68.98 ± 2.87	68.45 ± 1.76	82.25 ± 1.84	42.71 ± 1.74	71.68 ± 2.30	55.42 ± 0.32	76.55 ± 1.62	54.79 ± 2.73	78.20 ± 2.39
SURE	78.11 ± 3.91	88.61 ± 2.50	51.50 ± 2.74	74.32 ± 2.56	73.23 ± 1.34	83.72 ± 1.70	53.31 ± 1.26	**78.21 ± 1.74**	65.04 ± 2.71	82.65 ± 2.11	64.43 ± 1.59	83.51 ± 1.81
SURE-PFNL	39.26 ± 2.31	70.85 ± 2.21	42.30 ± 0.28	71.33 ± 1.24	56.73 ± 1.17	77.68 ± 1.50	50.60 ± 3.35	76.92 ± 2.60	46.38 ± 0.68	73.55 ± 1.34	59.34 ± 2.74	80.17 ± 2.41
SURE-RFUTE	78.93 ± 3.13	89.10 ± 2.21	49.33 ± 1.83	73.26 ± 2.02	69.18 ± 3.78	83.00 ± 2.64	52.21 ± 0.27	77.40 ± 1.62	65.38 ± 5.32	83.10 ± 2.91	62.12 ± 1.14	82.98 ± 1.49
SURE-SAUTE	72.30 ± 2.48	84.22 ± 2.44	46.25 ± 3.26	69.91 ± 2.81	67.22 ± 1.79	80.30 ± 1.97	**54.34 ± 2.35**	78.10 ± 2.51	66.68 ± 2.23	82.04 ± 2.10	60.13 ± 3.31	81.80 ± 2.70
SURE-CENDA	78.98 ± 2.31	88.23 ± 1.99	50.25 ± 2.23	73.92 ± 2.01	70.90 ± 2.45	83.38 ± 2.32	53.24 ± 3.64	77.70 ± 3.01	47.38 ± 1.35	75.13 ± 1.82	63.03 ± 0.62	82.71 ± 0.89
CLASS	**82.09 ± 1.16**	**91.28 ± 1.10**	**55.49 ± 3.93**	**77.13 ± 3.14**	**77.44 ± 1.29**	**87.44 ± 1.31**	53.37 ± 1.47	77.88 ± 1.52	**67.86 ± 1.04**	**85.32 ± 1.11**	**68.42 ± 2.98**	**86.99 ± 2.50**
win/tie/loss	20/0/0	20/0/0	20/0/0	20/0/0	20/0/0	20/0/0	18/0/2	17/1/2	20/0/0	20/0/0	20/0/0	20/0/0

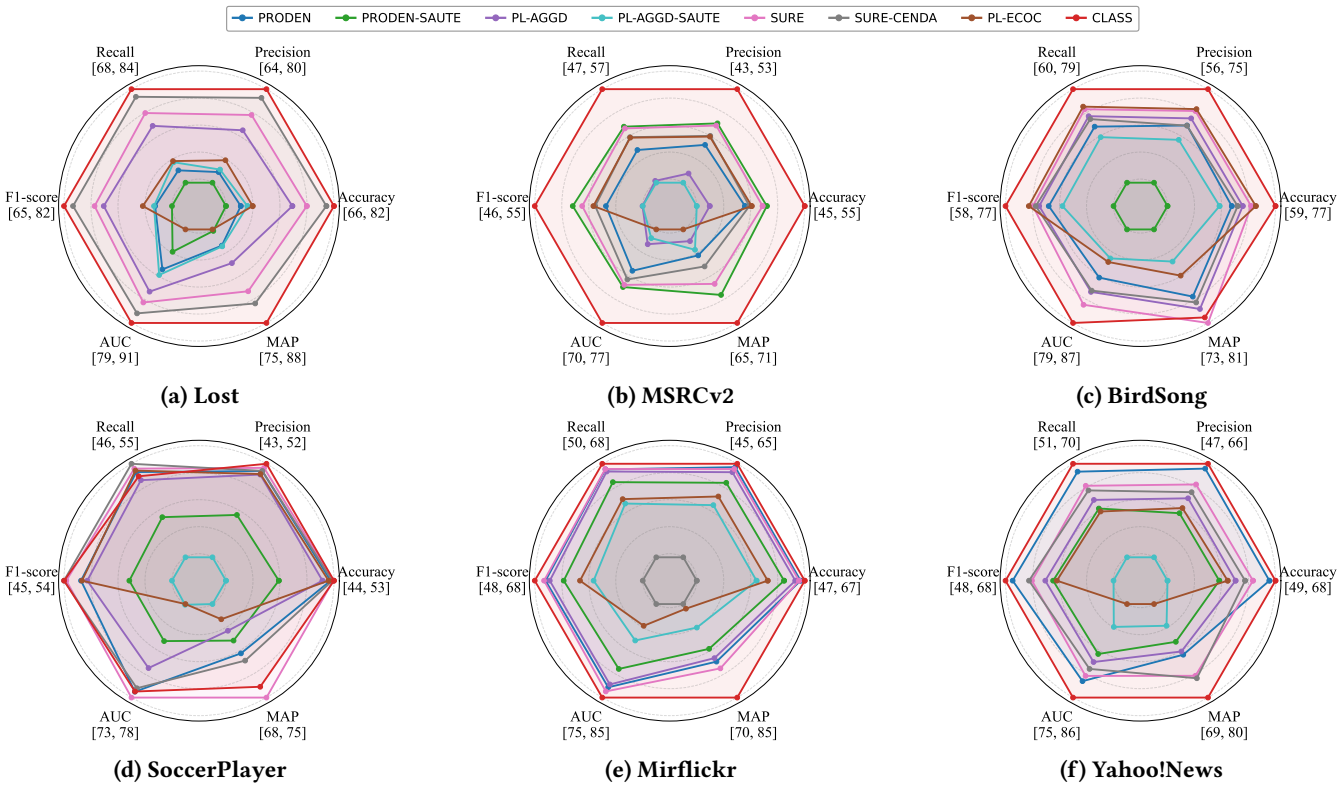

Figure 3: Comparison of all methods under multiple metrics on six real-world partial label datasets.

Lost, MSRCv2, BirdSong, SoccerPlayer, Mirflickr and Yahoo!News. Meanwhile, six classification metrics: ACC, AUC, MAP, F1, Precision and Recall are used to evaluate their performance. The results of ACC and AUC are reported in Tables 1. Besides, to deeply compare the comprehensive performance of each method, Figure 3 presents a radar chart visualization comparing CLASS against seven top-performing baselines: PL-ECOC, PRODEN, PRODEN-SAUTE, PL-AGGD, PL-AGGD-SAUTE, SURE, and SURE-CENDA, across six evaluation metrics. From the reported results, we can observe that:

- Compared with existing SOTA baselines, CLASS consistently outperforms other methods across multiple PLL datasets in most cases. For example, as shown in Figure 3 (b), CLASS forms the largest envelope range across six metrics on MSRCv2.

Table 2: Accuracy and AUC performance on four synthetic partial label datasets under different FPLs

FPLs	Method	MFEAT		HAR		Isolet		Dermatology	
		ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
$r = 1$	PL-ECOC	85.09 ± 3.21	87.30 ± 3.20	97.25 ± 1.89	97.50 ± 1.80	96.05 ± 2.10	96.10 ± 2.10	95.00 ± 2.85	95.40 ± 2.60
	PL-ECOC-PFNL	84.58 ± 3.42	86.50 ± 3.15	96.80 ± 1.78	97.20 ± 1.85	95.54 ± 2.05	95.20 ± 2.25	94.56 ± 2.78	94.29 ± 2.90
	PL-ECOC-RFUTE	90.12 ± 1.98	91.15 ± 2.10	98.50 ± 1.25	98.50 ± 1.25	96.82 ± 1.40	97.15 ± 1.35	96.50 ± 2.15	96.85 ± 2.10
	PL-ECOC-SAUTE	91.80 ± 1.75	92.90 ± 1.80	98.80 ± 1.05	99.10 ± 1.10	97.08 ± 1.20	97.25 ± 1.10	96.00 ± 1.65	97.10 ± 1.80
	PL-ECOC-CENDA	92.01 ± 1.68	93.10 ± 1.90	98.90 ± 0.98	99.15 ± 1.05	97.20 ± 1.10	96.40 ± 1.10	97.50 ± 1.48	97.30 ± 1.60
	PRODEN	84.62 ± 3.19	87.10 ± 3.25	97.42 ± 1.95	97.60 ± 1.90	95.83 ± 2.12	96.15 ± 2.05	94.87 ± 2.88	95.25 ± 2.80
	PRODEN-PFNL	85.35 ± 3.38	87.30 ± 3.10	97.00 ± 1.88	97.50 ± 1.85	95.35 ± 2.03	95.40 ± 2.15	94.30 ± 2.73	94.79 ± 2.75
	PRODEN-RFUTE	89.13 ± 2.45	90.05 ± 2.50	98.20 ± 1.68	98.60 ± 1.35	96.53 ± 1.83	97.05 ± 1.90	96.20 ± 2.42	96.60 ± 2.40
	PRODEN-SAUTE	91.17 ± 1.92	91.05 ± 2.00	98.60 ± 1.35	98.85 ± 1.45	96.77 ± 1.50	97.33 ± 1.50	96.88 ± 2.18	97.05 ± 2.00
	PRODEN-CENDA	91.93 ± 1.78	92.20 ± 1.90	98.80 ± 1.02	98.90 ± 1.00	97.04 ± 1.17	97.55 ± 1.20	97.20 ± 1.55	97.15 ± 1.70
	PL-AGGD	85.75 ± 3.12	87.45 ± 3.25	98.93 ± 0.89	98.85 ± 0.80	96.66 ± 1.08	96.86 ± 1.00	90.41 ± 3.85	91.10 ± 3.85
	PL-AGGD-PFNL	86.24 ± 3.28	87.55 ± 3.30	98.50 ± 0.95	98.70 ± 0.90	96.30 ± 1.05	96.60 ± 1.05	93.04 ± 3.78	93.14 ± 3.90
	PL-AGGD-RFUTE	90.36 ± 2.05	91.45 ± 2.20	98.70 ± 0.78	98.80 ± 0.75	96.46 ± 1.02	96.75 ± 1.10	93.50 ± 3.65	93.85 ± 3.80
	PL-AGGD-SAUTE	91.75 ± 1.68	92.42 ± 1.76	98.90 ± 0.65	98.80 ± 0.70	96.50 ± 1.00	96.78 ± 0.98	93.85 ± 3.58	93.52 ± 3.58
	PL-AGGD-CENDA	92.76 ± 1.55	93.20 ± 1.60	99.00 ± 0.50	99.00 ± 0.50	96.60 ± 0.98	97.00 ± 1.05	94.00 ± 3.45	94.30 ± 3.60
	SURE	88.49 ± 2.85	89.05 ± 2.85	98.85 ± 1.20	98.90 ± 1.30	95.00 ± 2.35	95.29 ± 2.45	93.00 ± 3.15	93.40 ± 3.20
	SURE-PFNL	87.92 ± 2.78	88.30 ± 2.90	98.40 ± 1.45	98.70 ± 1.40	94.57 ± 2.58	94.80 ± 2.20	92.56 ± 3.08	93.15 ± 3.25
	SURE-RFUTE	90.81 ± 2.35	90.35 ± 2.50	98.60 ± 1.12	98.75 ± 1.20	95.53 ± 2.25	95.66 ± 2.10	93.54 ± 3.02	94.45 ± 3.00
	SURE-SAUTE	91.74 ± 1.88	91.30 ± 2.10	98.80 ± 1.08	98.95 ± 1.10	95.82 ± 2.12	95.97 ± 2.00	93.87 ± 2.95	94.65 ± 3.00
	SURE-CENDA	92.40 ± 1.75	92.60 ± 1.80	98.90 ± 0.95	99.00 ± 0.95	96.08 ± 2.05	96.13 ± 2.10	94.07 ± 2.88	94.90 ± 2.90
	CLASS	**95.30 ± 0.15**	**95.15 ± 0.12**	**99.85 ± 0.10**	**99.90 ± 0.08**	**97.18 ± 0.11**	**97.60 ± 0.09**	**98.36 ± 0.10**	**98.42 ± 0.08**
$r = 2$	PL-ECOC	86.59 ± 3.15	88.72 ± 3.05	96.53 ± 1.92	95.87 ± 1.88	95.52 ± 2.15	94.81 ± 2.03	94.00 ± 2.90	92.94 ± 2.65
	PL-ECOC-PFNL	87.42 ± 3.35	89.53 ± 3.13	96.00 ± 1.82	96.27 ± 1.76	95.06 ± 2.08	95.13 ± 1.91	93.50 ± 2.82	93.43 ± 2.51
	PL-ECOC-RFUTE	88.30 ± 2.05	90.92 ± 2.05	97.80 ± 1.30	96.83 ± 1.29	96.33 ± 1.45	95.52 ± 1.62	95.50 ± 2.20	94.34 ± 2.21
	PL-ECOC-SAUTE	91.27 ± 1.82	92.12 ± 1.81	98.04 ± 1.10	97.32 ± 1.02	96.59 ± 1.25	95.32 ± 1.14	96.00 ± 1.70	94.54 ± 1.81
	PL-ECOC-CENDA	92.29 ± 1.72	92.82 ± 1.60	98.20 ± 1.02	97.53 ± 0.97	96.73 ± 1.15	95.53 ± 1.11	96.50 ± 1.52	94.72 ± 1.41
	PRODEN	85.39 ± 3.22	88.12 ± 3.12	96.73 ± 2.00	96.85 ± 1.87	95.37 ± 2.17	94.95 ± 2.11	93.80 ± 2.92	93.52 ± 2.73
	PRODEN-PFNL	86.75 ± 3.31	88.98 ± 3.08	96.34 ± 1.93	96.58 ± 1.74	94.83 ± 2.08	95.24 ± 2.01	93.30 ± 2.78	93.18 ± 2.63
	PRODEN-RFUTE	88.18 ± 2.52	90.38 ± 2.14	97.50 ± 1.73	97.02 ± 1.33	96.05 ± 1.88	95.37 ± 1.58	95.20 ± 2.47	94.21 ± 2.25
	PRODEN-SAUTE	89.50 ± 2.05	91.60 ± 1.81	97.85 ± 1.40	97.42 ± 1.05	96.22 ± 1.55	95.51 ± 1.31	95.80 ± 2.23	94.38 ± 2.14
	PRODEN-CENDA	92.15 ± 1.82	92.42 ± 1.64	98.07 ± 1.07	97.63 ± 0.94	96.57 ± 1.22	95.56 ± 1.13	96.20 ± 1.60	94.47 ± 1.44
	PL-AGGD	86.25 ± 3.17	88.62 ± 2.92	98.54 ± 0.92	97.29 ± 1.19	96.54 ± 1.12	95.25 ± 1.38	94.52 ± 2.90	94.08 ± 2.53
	PL-AGGD-PFNL	87.91 ± 3.21	89.30 ± 2.85	98.10 ± 0.98	97.43 ± 1.08	96.24 ± 1.08	95.35 ± 1.28	94.00 ± 2.85	94.14 ± 2.31
	PL-AGGD-RFUTE	88.83 ± 2.12	90.68 ± 2.05	98.30 ± 0.82	97.85 ± 0.98	96.32 ± 1.05	95.46 ± 1.22	94.50 ± 2.78	94.30 ± 2.15
	PL-AGGD-SAUTE	90.56 ± 1.75	91.23 ± 1.75	98.56 ± 0.70	97.94 ± 0.88	96.45 ± 1.03	95.55 ± 1.06	94.70 ± 2.65	94.42 ± 2.05
	PL-AGGD-CENDA	92.90 ± 1.58	92.37 ± 1.58	98.60 ± 0.55	98.01 ± 0.77	96.57 ± 1.02	95.70 ± 1.08	94.80 ± 2.58	94.56 ± 2.18
	SURE	85.71 ± 2.92	87.18 ± 3.00	98.10 ± 1.25	96.92 ± 1.77	94.50 ± 2.40	94.92 ± 2.04	92.00 ± 3.20	93.25 ± 2.58
	SURE-PFNL	87.37 ± 2.85	88.06 ± 2.92	97.70 ± 1.50	97.12 ± 1.60	94.04 ± 2.63	95.11 ± 2.12	91.50 ± 3.13	93.04 ± 2.60
	SURE-RFUTE	88.56 ± 2.42	89.25 ± 2.20	97.90 ± 1.17	97.38 ± 1.25	95.03 ± 2.30	95.36 ± 1.58	92.50 ± 3.07	93.17 ± 2.30
	SURE-SAUTE	89.47 ± 1.95	90.33 ± 2.10	98.13 ± 1.13	97.48 ± 1.05	95.32 ± 2.17	95.46 ± 1.28	92.80 ± 3.00	93.33 ± 2.12
	SURE-CENDA	92.81 ± 1.82	93.12 ± 1.70	98.20 ± 1.00	97.60 ± 0.98	95.58 ± 2.10	96.57 ± 1.18	93.00 ± 2.92	93.46 ± 2.02
	CLASS	**94.60 ± 0.18**	**94.62 ± 0.19**	**98.95 ± 0.12**	**98.97 ± 0.12**	**96.76 ± 0.13**	**96.75 ± 0.14**	**96.71 ± 0.17**	**96.73 ± 0.17**
$r = 3$	PL-ECOC	84.87 ± 3.33	85.02 ± 3.27	95.78 ± 1.92	95.52 ± 1.83	94.98 ± 2.23	94.28 ± 2.21	93.02 ± 2.91	92.95 ± 3.10
	PL-ECOC-PFNL	85.70 ± 3.26	85.82 ± 3.22	95.18 ± 1.86	95.76 ± 1.73	94.54 ± 2.11	94.51 ± 2.09	92.47 ± 2.88	93.12 ± 2.88
	PL-ECOC-RFUTE	88.21 ± 2.13	87.13 ± 2.17	96.98 ± 1.33	96.04 ± 1.38	95.78 ± 1.52	95.03 ± 1.59	94.47 ± 2.22	93.53 ± 2.33
	PL-ECOC-SAUTE	88.94 ± 1.87	87.57 ± 1.79	97.23 ± 1.13	96.14 ± 1.02	96.03 ± 1.31	95.26 ± 1.28	95.02 ± 1.76	93.35 ± 1.77
	PL-ECOC-CENDA	88.52 ± 1.81	87.81 ± 1.73	97.52 ± 1.07	96.24 ± 0.98	96.19 ± 1.19	95.42 ± 1.29	95.51 ± 1.56	93.56 ± 1.62
	PRODEN	83.54 ± 3.30	84.19 ± 3.39	96.03 ± 2.03	94.98 ± 2.05	94.82 ± 2.20	94.19 ± 2.13	92.78 ± 2.98	92.48 ± 3.09
	PRODEN-PFNL	86.68 ± 3.27	85.41 ± 3.28	95.58 ± 2.00	95.43 ± 1.91	94.32 ± 2.12	94.44 ± 2.04	92.28 ± 2.85	92.74 ± 2.95
	PRODEN-RFUTE	88.30 ± 2.57	87.08 ± 2.59	96.81 ± 1.80	96.09 ± 1.78	95.52 ± 1.91	95.16 ± 1.90	94.18 ± 2.54	93.14 ± 2.65
	PRODEN-SAUTE	88.56 ± 2.16	87.48 ± 2.35	97.02 ± 1.47	96.15 ± 1.60	95.72 ± 1.62	95.31 ± 1.71	94.81 ± 2.30	93.28 ± 2.43
	PRODEN-CENDA	88.66 ± 1.87	87.81 ± 2.08	97.32 ± 1.14	96.26 ± 1.49	95.97 ± 1.26	95.45 ± 1.63	95.23 ± 1.66	93.57 ± 2.18
	PL-AGGD	85.24 ± 3.25	85.91 ± 3.29	98.02 ± 0.96	96.43 ± 1.09	96.40 ± 1.17	94.94 ± 1.31	91.80 ± 3.07	92.74 ± 2.99
	PL-AGGD-PFNL	86.46 ± 3.13	86.97 ± 3.13	97.63 ± 1.04	96.65 ± 1.02	96.09 ± 1.11	95.06 ± 1.41	91.98 ± 2.90	92.86 ± 2.88
	PL-AGGD-RFUTE	88.31 ± 2.16	87.51 ± 2.20	97.82 ± 0.87	96.94 ± 1.03	96.21 ± 1.05	95.19 ± 1.27	92.49 ± 2.87	93.07 ± 2.72
	PL-AGGD-SAUTE	88.87 ± 1.85	87.75 ± 1.91	98.00 ± 0.74	97.05 ± 0.94	96.33 ± 1.08	95.41 ± 1.06	92.78 ± 2.71	93.18 ± 2.47
	PL-AGGD-CENDA	88.74 ± 1.67	88.02 ± 1.77	97.22 ± 0.62	97.15 ± 0.89	96.42 ± 1.06	95.52 ± 1.02	93.01 ± 2.63	93.29 ± 2.33
	SURE	84.18 ± 2.95	84.60 ± 2.94	97.39 ± 1.29	95.59 ± 1.31	94.02 ± 2.47	93.38 ± 2.30	91.03 ± 3.22	91.61 ± 2.98
	SURE-PFNL	86.27 ± 2.94	85.65 ± 2.83	97.01 ± 1.53	95.74 ± 1.18	93.53 ± 2.66	93.53 ± 2.27	90.48 ± 3.20	91.73 ± 2.87
	SURE-RFUTE	86.69 ± 2.46	85.92 ± 2.53	97.22 ± 1.25	95.85 ± 1.11	94.49 ± 2.34	93.76 ± 2.14	91.52 ± 3.14	91.83 ± 2.78
	SURE-SAUTE	88.56 ± 2.05	87.14 ± 2.12	97.42 ± 1.20	96.05 ± 1.07	94.81 ± 2.20	94.07 ± 2.08	91.79 ± 3.03	92.02 ± 2.64
	SURE-CENDA	89.33 ± 1.91	87.74 ± 1.84	97.49 ± 1.06	96.16 ± 0.97	95.01 ± 2.14	94.28 ± 1.83	91.99 ± 2.95	92.26 ± 2.56
	CLASS	**90.30 ± 0.28**	**91.78 ± 0.29**	**98.02 ± 0.15**	**98.17 ± 0.14**	**96.48 ± 0.16**	**96.65 ± 0.15**	**96.14 ± 0.25**	**95.42 ± 0.27**
	win/tie/loss	60/0/0	60/0/0	60/0/0	60/0/0	60/0/0	60/0/0	60/0/0	60/0/0

The reason may be that our proposed CLASS designs a unified mechanism that jointly optimizes feature representation and label disambiguation for PLL.

- It is obvious that among other baselines, SURE and its variants perform better than others in most scenarios. It may be caused by the introduction of a maximum infinity norm

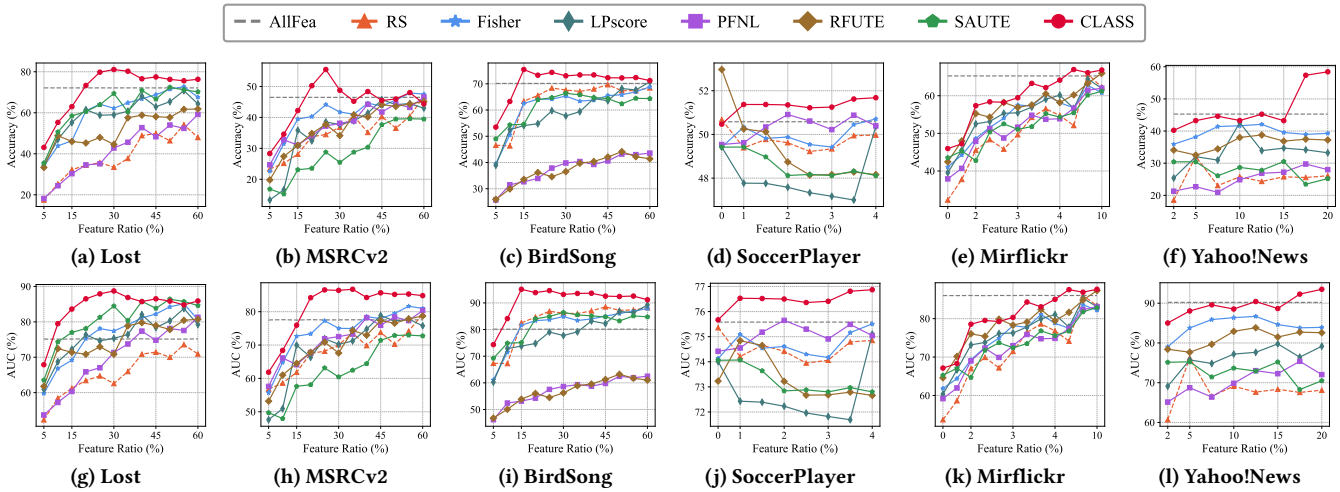

Figure 4: Comparison of classification Accuracy and AUC under varying feature retention ratios on six real-world partial label datasets. All methods use PL-AGGD as the base classifier.

regularization in partial label learning, which effectively enhances the mutual exclusivity among candidate labels.

Sensitivity to False Positive Labels:

To further evaluate the sensitivity of CLASS to false positive labels (FPLs), we perform experiments on synthetic partial label datasets: MFEAT, HAR, Isolet and Dermatology, which were constructed using several publicly available high-dimensional multi-class datasets from UCI. For each instance, to control the ratios of FPLs, we keep the the ground-truth label and randomly generate and add $r \in \{1, 2, 3\}$ FPLs to form partial label datasets. Besides, we evaluate the performance of compared methods with the same metrics: ACC, AUC, MAP, F1, Precision and Recall. The results of ACC and AUC under partial label ratios $r = 1$ and $r = 2$ are reported in Tables 2. By analyzing these results, we can observe that:

- The performance of our CLASS is still better than other methods in most cases under different number of FPLs. For example, as shown by the t-test results, the numbers of win/tie/loss of our methods are 40, 0 and 0, respectively. The reason may be that CLASS designs a cluster-guided disambiguation module to dynamically align predicted labels with the semantics of prototypes in each class.
- In addition, the proposed method demonstrates strong robustness under varying degrees of FPLs. With the number of FPLs increases, its performance do not show significant decline, while the performance of SURE-SAUTE and SURE-CENDA exhibits noticeable fluctuations. This indicates that our method has strong adaptability and resistance to FPLs.

4.3 Feature Selection Performance

To verify the effectiveness of CLASS in feature selection, we compare it with two traditional feature selection methods: Fisher score (Fisher) and Laplacian score (LPscore), and partial lable methods PFNL, RFUTE and SAUTE, which are using the same base classifier PL-AGGD. Meanwhile, we also take PL-AGGD on all features as a baseline AllFea and evaluate their performance with the same six metrics. The results of ACC and AUC of compared methods under different ratios of features on Lost, MSRCv2, BirdSong, SoccerPlayer, Mirflickr and Yahoo!News are shown in Figure 4. From these results, we observe that:

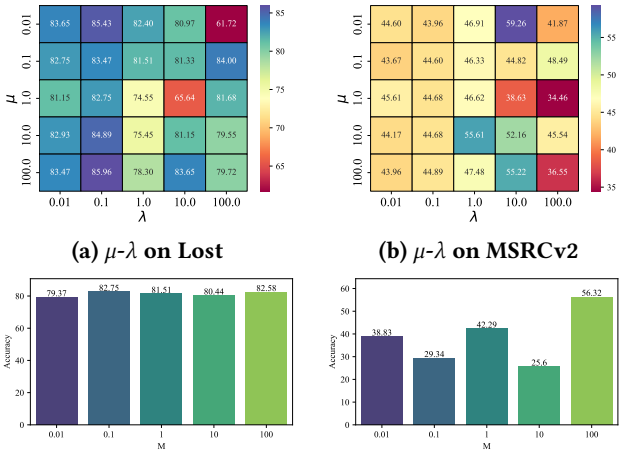

Figure 5: Sensitivity Analysis of Parameters μ, λ, and M for the CLASS Method on Lost and MSRCv2 Datasets

- CLASS consistently achieves highest accuracy under different feature ratios. Besides, the advantage of our method is more remarkable if we have a smaller feature ratios. It may be caused by integrating deep neural networks with a linear sparse gating mechanism to capture high-order semantic relations and feature interactions, which promote selecting more discriminative features.
- It is obvious that all compared methods perform better with the increase of feature ratio in almost all cases, since we have more feature information with large feature ratio.
- Generally, AllFea performs best in all the baseline methods in most cases since it utilizes data points with all features.

Nevertheless, our CLASS still perform better than it since we reduce the feature redundancy by selection.

4.4 Parameter Sensitivity

We have three hyper-parameters in our model, i.e., λ, μ and M. In this subsection, we analyze their impact on CLASS by tuning each within $\{0.01, 0.1, 1, 10, 100\}$, with results on Lost and MSRCv2 shown in Figure 5. For λ and μ, as shown in Figure 5(a)-(b), both datasets exhibit lower sensitivity to μ but higher sensitivity to λ. On the Lost dataset, better performance is observed with mild λ values. For MSRCv2, optimal results are achieved with larger λ. Fixed other parameters,as shown in Figure 5(c)-(d), M is insensitive to Lost with stable accuracy while larger M suits MSRCv2.

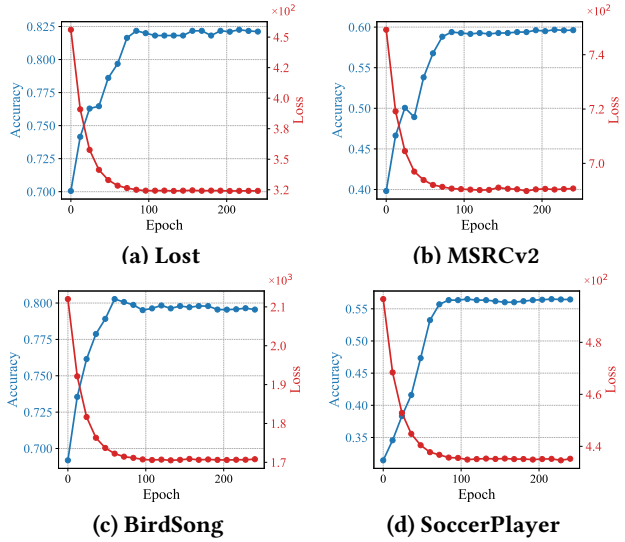

Figure 6: Accuracy and training loss of the proposed CLASS over training epochs on several partial label datasets.

4.5 Convergence Analysis

To better understand the convergence behaviour of our method, we recorded the changes in classification accuracy and training loss during the training of CLASS on Lost, MSRCv2, BirdSong and SoccerPlayer in Figure 6. The blue line represents accuracy and the red line represents loss over training epochs. The results show that the loss drops quickly within the first 50 to 100 epochs and then becomes stable, while the accuracy gradually increases and eventually levels off. The training process is smooth, without significant fluctuations. This consistent trend of rapid initial convergence followed by stable performance holds across all tested datasets, demonstrating that CLASS maintains reliable training dynamics regardless of data complexity. This validates the effectiveness of its optimization algorithm.

5 Conclusion

To simultaneously tackle the problems of feature selection and label disambiguation, we propose a novel partial label learning method CLASS in this paper. Specifically, CLASS integrates nonlinear deep networks with linear structured sparsity constraints to capture the high-order semantic interactions of features and exploit the underlying nonlinear structured information. Meanwhile, a cluster-guided disambiguation strategy is designed to effectively mine the semantic structure among candidate labels and reduce label disambiguation of partial data. Moreover, we develop an efficient alternative optimization algorithm to solve the formulated objective. Comprehensive experimental results demonstrate that CLASS consistently outperforms state-of-the-art PLL methods. Besides, extending it to tackle the partial label data with temporal dependency is a promising direction of future.

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A Notations

Given a d-dimensional feature space $\mathcal{X} = \mathbb{R}^d$ and label space $\mathcal{Y} = \{y_1, ..., y_q\}$, the training set is $\mathcal{D} = \{(\mathbf{x}_i, S_i)\}_{i=1}^m$ where $\mathbf{x}_i \in \mathbb{R}^d$ and $S_i \subseteq \mathcal{Y}$, which containing the true label $\mathbf{y}_i$. To reduce noise, we select $d' = \rho d$ features to form $\tilde{\mathbf{x}}_i \in \mathbb{R}^{d'}$. The partial label learning goal is to train a classifier $f : \mathbb{R}^{d'} \to \mathcal{Y}$ that predicts a single label for unseen instances from $\{(\tilde{\mathbf{x}}_i, S_i)\}$.

B Solution of Proposed CLASS

B.1 Optimization

Equation (9) can be formulated as:

$$\mathcal{L}(\mathbf{f}_i, \alpha, \boldsymbol{\beta}_i) = \|\mathbf{f}_i + \frac{1}{2}\boldsymbol{\psi}_i\|^2 - \alpha(\mathbf{f}_i^\top \mathbf{1} - 1) - \boldsymbol{\beta}_i^\top \mathbf{f}_i, \tag{17}$$

where $\boldsymbol{\psi}_i = [\psi_{i1}, \ldots, \psi_{iq}]^\top$, $\alpha \in \mathbb{R}$ and $\boldsymbol{\beta}_i \in \mathbb{R}^q_{\geq 0}$ are Lagrange multipliers. According to the KKT conditions [28], we have

$$\nabla_{\mathbf{f}_i}\mathcal{L} = 2(f_i + \frac{1}{2}\boldsymbol{\psi}_i) - \alpha 1 - \boldsymbol{\beta}_i = 0, \tag{18}$$

we first shift the terms and provide the implicit expression of the optimal solution of f_{ic} is

$$\mathbf{f}_i = \frac{1}{2}\left(\alpha\mathbf{1} + \boldsymbol{\beta}_i - \boldsymbol{\psi}_i\right), \tag{19}$$

combining the complementary slack condition $\boldsymbol{\beta}_i^\top \mathbf{f}_i = 0$ with $\boldsymbol{\beta}_i \geq \mathbf{0}$, we obtain the following:

$$\begin{cases} f_{ic} = \frac{1}{2}(-\psi_{ic} + \alpha), & \text{if } \psi_{ic} < \alpha, \\ f_{ic} = 0, & \text{if } \psi_{ic} \geq \alpha, \end{cases} \tag{20}$$

integrating the above conditions, the optimal solution can be expressed as:

$$f_{ic} = \left(-\frac{1}{2}\psi_{ic} + \frac{\alpha}{2}\right)_+, \tag{21}$$

where $(\cdot)_+$ denotes the positive part. Substituting into the constraint $\sum_{c=1}^{q} f_{ic} = 1$, we can get the optimal α:

$$\sum_{c=1}^{q}\left(-\frac{1}{2}\psi_{ic}+\frac{\alpha}{2}\right)=1 \implies \alpha=\frac{2}{q}+\frac{1}{q}\sum_{c=1}^{q}\psi_{ic}. \tag{22}$$

B.2 Warm Start

Since the training set $D = \{(\mathbf{x_i}, S_i)|1 \le i \le m\}$ in the initial stage contains a large number of false positive labels, if D is directly used to calculate the class centers $\mathbf{V} = \{\mathbf{v_c}|1 \le c \le q\}$, an excessive amount of noise will be introduced. Therefore, before solving the objective function, the variables need to be reasonably initialized.

B.2.1 Initialize $\mathbf{F}$. Construct a weighted directed graph structure $\mathcal{G} = \{\mathbf{T}, \mathbf{E}, \mathbf{S}\}$, where $\mathbf{T} = \{\mathbf{x_i}|1 \le i \le m\}$ represents the vertex set, $\mathbf{E} = \{(\mathbf{x_i}, \mathbf{x_j})|1 \le i, j \le m, i \ne j\}$ represents the edge set, and $\mathbf{S} = [S_{ij}]_{m\times m}$ denotes the connection relationship of the graph with values ranging from $[0, 1]$. A larger value indicates a higher similarity between two vertices. $\mathcal{N}(\mathbf{x_i})$ denotes the set of k nearest neighbors of sample $\mathbf{x_i}$ in the partial label dataset D. The similarity matrix S between $\mathbf{x_i}$ and $\mathbf{x_j}$ is calculated using the Gaussian kernel:

$$S_{ij}=\begin{cases}\exp(-\frac{\|\mathbf{x_i}-\mathbf{x_j}\|^2}{2\sigma^2}), & if \quad \mathbf{x_j}\in\mathcal{N}(\mathbf{x_i})\\ 0, & otherwise\end{cases} \tag{23}$$

Define the label confidence matrix as $\mathbf{F} = [f_{ic}]_{m\times q} \in \mathbb{R}^{m\times q}$, where f_{ic} represents the confidence that $\mathbf{x_i}$ belongs to class c. Initialize $\mathbf{F}^{(0)}$ as follows:

$$\forall 1\le i\le m: f_{ic}^{(0)}=\begin{cases}\frac{1}{|\mathbf{y_i}|}, & if \quad y_{i,c}=1\\ 0, & otherwise\end{cases} \tag{24}$$

Among them, $|\mathbf{y_i}|$ denotes the total number of candidate labels for $\mathbf{x_i}$. According to the label propagation process[35, 50], $\mathbf{F}$ is updated as follows:

$$\mathbf{F}^{(t)} = \gamma\cdot\mathbf{H}^{(t-1)} + (1-\gamma)\cdot\mathbf{F}^{(0)} \tag{25}$$

Among them, $\gamma \in (0, 1)$ is the balancing parameter, and H is the standardized S, that is, $H = WD^{-1}$, where $D = \mathrm{diag}[d_1, d_2, \ldots, d_m]$, and the diagonal element $d_j = \sum_{i=1}^{m} S_{ij}$ represents the sum of the similarities of each sample $\mathbf{x_i}$. When $\mathbf{F}$ converges or reaches the maximum number of iterations, it is denoted as $\mathbf{F}^* = [f_{ic}^*]_{m\times q}$. Further, $\mathbf{F}^*$ is normalized row by row to $\hat{F} = [\hat{f}_{ic}]_{m\times q}$:

$$\forall 1\le i\le m: \hat{f}_{ic}=\begin{cases}\frac{f_{ic}^*}{\sum_{l=1}^{q} f_{il}^* y_{il}}, & if \quad y_{ic}=1\\ 0, & otherwise\end{cases} \tag{26}$$

B.2.2 Initialize v_c. The class with the highest confidence of the sample $\mathbf{x_i}$ is determined as its initial ground-truth label $\mathbf{y_i}^{(0)}$:

$$\mathbf{y_i} = \arg\max_{c\in y} \hat{f}_{ic} \tag{27}$$

Based on the initial ground-truth label, the class center v_c is initialized in the following form:

$$\forall 1\le c\le q: \quad \mathbf{v_c}=\frac{\sum_{i=1}^{m}\mathbb{I}(\hat{\mathbf{y}}_\mathbf{i}=c)\cdot\mathbf{x}_i}{\sum_{i=1}^{m}\mathbb{I}(\hat{\mathbf{y}}_\mathbf{i}=c)} \tag{28}$$

Among them, $\mathbb{I}(\cdot)$ is an indicator function, which takes the value of 1 when the condition inside the parentheses is satisfied and takes the value of 0 when the condition is not satisfied.

B.2.3 Initialize $\mathbf{\Theta}$ *and* $\mathbf{W}$. The weight matrix $\mathbf{\Theta} \in \mathbb{R}^{d\times c}$ of the linear part is responsible for feature selection and extraction of linear relationships of the input. We have found that the generalization performance of the method from dense to sparse for $\mathbf{\Theta}$ is significantly better than that of the methods from sparse to dense. Among them, the methods from sparse to dense is prone to fall into a local minimum of formulated objective. Therefore, at the initial stage, $\mathbf{\Theta}$ can be initialized as

$$\mathbf{\Theta}_0=\begin{pmatrix}1 & 1 & \cdots & 1\\ 1 & 1 & \cdots & 1\\ \vdots & \vdots & \ddots & \vdots\\ 1 & 1 & \cdots & 1\end{pmatrix}.$$

The nonlinear part is extracted by a neural network. Among them, the weight $\mathbf{W}^{(1)} \in \mathbb{R}^{d\times q}$ of the first hidden layer is controlled by $\mathbf{\Theta}$, and it is an important variable for maintaining the feature consistency between the linear part and the nonlinear part, which is initialized through the backpropagation of the network.

C Experimental Details

C.1 Detailed Description of Datasets

The six real-world datasets used in this experiment include:

- **Lost**, **SoccerPlayer**, and **Yahoo! News**: Derived from automatic face naming tasks. Each instance corresponds to a cropped face image, and candidate labels are extracted from associated image captions or surrounding text.
- **MSRCv2**: An object classification dataset where each instance is an image region, and co-occurring object tags are treated as candidate labels.
- **BirdSong**: Involves bird call classification. Each instance is an audio syllable, with candidate labels being the bird species likely to vocalize within the same 10-second recording.
- **Mirflickr**: A web image classification dataset where each image is associated with candidate labels extracted from surrounding webpage keywords.

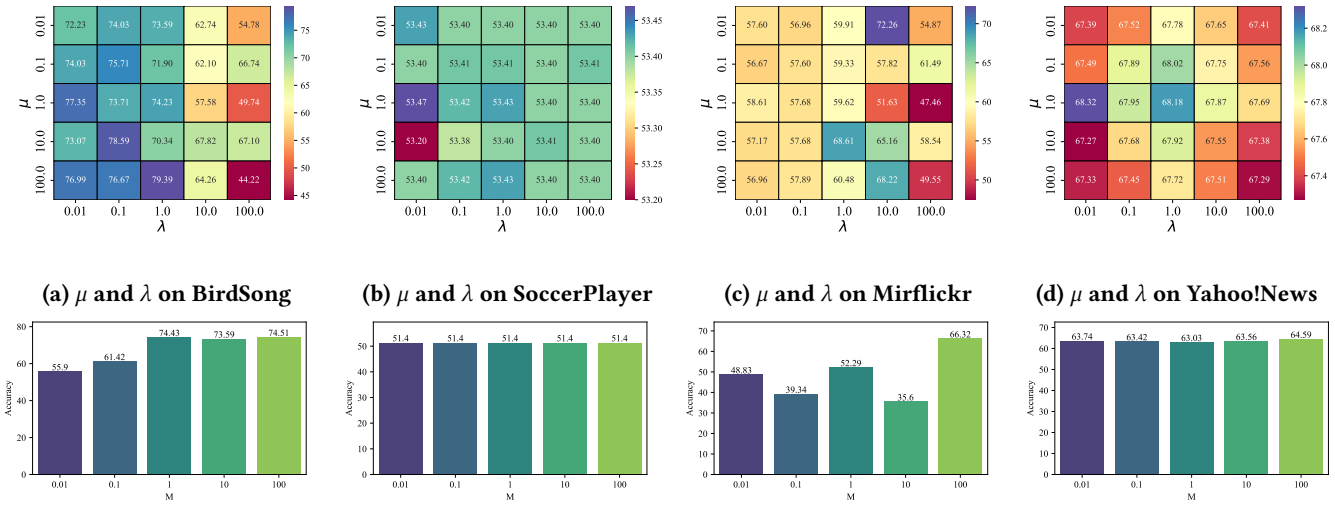
While **the four synthetic partial label datasets** include [1]: **MFEAT**, **HAR**, **Isolet**, and **Dermatology**. These datasets differ significantly in terms of the number of instances, feature dimensionality, and class cardinality, providing a diverse testing ground to evaluate the adaptability and effectiveness of the proposed method under various structural conditions.

C.2 Detailed Description of Baselines

Partial label classification methods include:

- **PL-ECOC** [49]: Avoids disambiguation by decomposing the multi-class task into multiple binary classification problems via error-correcting output codes.
- **PL-AGGD** [36]: Utilizes graph structures to refine soft labels by enforcing local smoothness.
- **PRODEN** [21]: Progressively updates label confidences through model feedback to improve disambiguation.

[1]All datasets are from the UCI Machine Learning Repository:
MFEAT: https://doi.org/10.24432/C5HC70;
HAR: https://doi.org/10.24432/C54S4K;
Isolet: https://doi.org/10.24432/C51G69;
Dermatology: https://doi.org/10.24432/C5FK5P.



(a) μ and λ on BirdSong (b) μ and λ on SoccerPlayer (c) μ and λ on Mirflickr (d) μ and λ on Yahoo!News

(e) M on BirdSong (f) M on SoccerPlayer (g) M on Mirflickr (h) M on Yahoo!News

Figure 7: Sensitivity Analysis of Parameters μ, λ, and M of our CLASS on BirdSong, SoccerPlayer, Mirflickr and Yahoo!News.

Table 3: Running time (in seconds) of different methods across datasets and feature ratios. The gray row indicates our proposed method CLASS. "–" denotes CENDA cannot reduce to the specified feature count and is thus omitted.

	Method	MSRCv2	BirdSong	Mirflickr	Yahoo!News
	FS Ratio	**0.10**	**0.10**	**0.01**	**0.01**
Classifier	**FS Count**	**5**	**4**	**15**	**2**
PL-ECOC	Fisher	0.94	0.65	12.94	13.95
	LPscore	0.66	0.66	14.44	15.31
	PFNL	2.83	1.85	37.65	40.96
	RFUTE	4.63	4.53	93.97	99.61
	SAUTE	7.21	6.02	638.76	152.14
	CENDA	2.84	2.87	258.34	54.69
PRODEN	Fisher	21.63	18.43	51.60	55.91
	LPscore	21.65	18.46	53.31	57.92
	PFNL	23.34	19.63	76.13	82.17
	RFUTE	25.85	22.53	132.14	141.92
	SAUTE	28.63	24.80	677.46	194.57
	CENDA	23.38	20.46	297.63	96.51
PL-AGGD	Fisher	13.80	7.28	31.02	37.99
	LPscore	13.63	7.22	33.94	39.43
	PFNL	15.14	8.21	56.27	64.14
	RFUTE	17.69	11.97	112.38	123.94
	SAUTE	20.38	13.34	657.63	176.11
	CENDA	15.96	9.80	277.57	78.68
SURE	Fisher	32.63	18.92	85.20	136.96
	LPscore	32.13	18.94	87.11	138.63
	PFNL	34.63	19.22	110.63	163.27
	RFUTE	36.28	22.43	166.46	222.34
	SAUTE	39.87	24.94	711.93	275.88
	CENDA	34.55	20.45	331.27	177.28
CLASS		15.08	7.29	30.00	60.59

- **SURE** [8]: Enhances supervision via pseudo-labeling and iterative retraining.

Feature selection methods include:

- **PFNL** [30]: Exploits manifold structure and predicted label distributions to score and rank features.
- **RFUTE** [33]: Applies ensemble-based decision trees to assess feature relevance on candidate labels.
- **SAUTE** [2]: Models feature selection as a submodular maximization problem that balances relevance and redundancy.
- **CENDA** [1]: Conducts dimensionality reduction by maximizing dependence between features and soft label confidences.

D Additional Results

D.1 Parameter Sensitivity Analysis

We analyze the impact of hyper-parameters λ, μ, and M on CLASS across BirdSong, SoccerPlayer, Mirflickr, and Yahoo!News, with each parameter tuned within $\{0.01, 0.1, 1, 10, 100\}$ in Figure 7. For λ and μ, all datasets show lower sensitivity to μ but varying sensitivity to λ. BirdSong shows low sensitivity to μ but better performance with mild λ values. SoccerPlayer remains nearly unchanged across all λ and μ settings. Mirflickr is more sensitive to λ, achieving optimal results with larger λ. Yahoo!News exhibits stable performance across λ and μ variations. Fixed other parameters, M shows distinct sensitivities. Yahoo and Soccer are insensitive to M with stable accuracy, Bird prefers smaller M, Mirflickr benefits from larger M.

D.2 Complexity Analysis

To evaluate the computational efficiency of different feature selection methods, we measured their runtime (in seconds) on real-world partial label datasets under varying feature selection ratios in Table 3. The results show that the proposed CLASS method maintains consistently low and stable runtime across all datasets and settings.